PROJECT PROPOSAL

Title: Flight Delay Analysis and Prediction using Machine Learning

1. Student Details

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2. Problem Statement

Flight delays are a major issue in the aviation industry, causing inconvenience to passengers and financial losses to airlines. The goal of this project is to analyze flight data and predict whether a flight will be delayed based on features such as airline, departure time, origin, destination, and distance. This helps improve scheduling efficiency and customer satisfaction.

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3. Domain

Data Science / Machine Learning (Aviation Analytics)

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4. Dataset Information

- Dataset Name: flights_sample_3m.csv
- Source: Kaggle
- Type: Structured tabular dataset
- Key Features:
    - Airline
    - Origin Airport
    - Destination Airport

- Departure Delay
- Arrival Delay
- Distance
- Flight Time Attributes

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5. Dataset Description

- **Source:** Kaggle
- **Dataset Name:** Flight Delay Analysis and Prediction using Machine Learning
- **Link:** Dataset Link: Dataset Link: https://www.kaggle.com/datasets/flight-delay-dataset

Dataset Features:

1. Airline

- Name or code of the airline operating the flight
- Example: AA, DL, UA

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2. Origin Airport

- Airport from where the flight starts
- Example: JFK, LAX, ORD

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3. Destination Airport

- Airport where the flight lands
- Example: SFO, ATL, MIA

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4. Departure Delay (dep_delay)

- Delay in flight departure (in minutes)

- Positive value → delayed
- Negative value → early departure

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5. Arrival Delay (arr_delay)

- Delay in flight arrival (in minutes)
- Used as target variable in prediction

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6. Distance

- Distance between origin and destination airports (in miles/km depending on dataset)
- Helps in analyzing delay patterns

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7. Time-related Features (if present)

- Month / Day / Hour of flight
- Used for trend analysis

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8. Categorical Features

- Airline
- Origin
- Destination

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9. Target Variable (for ML)

- **Arrival Delay OR Delayed (0/1 classification)**
- Used for prediction model

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6. Objectives

- To analyze flight dataset and understand key features like airline, origin, destination, and delays.
- To identify important factors affecting flight delays.
- To clean and preprocess the dataset by handling missing values and duplicates.
- To perform Exploratory Data Analysis (EDA) using different graphs and visualizations.
- To study relationships between variables such as distance and delay.
- To build a machine learning model to predict flight delays.
- To classify flights as delayed or on-time based on given features.
- To evaluate the performance of the model using accuracy and error metrics.
- To improve decision-making for airlines and passengers using data insights.

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7. Methodology

The project will follow these steps:

1. Data Collection

- Dataset is loaded from Kaggle / CSV file
- Large-scale flight records are used for analysis

2. Data Preprocessing

- Handle missing values
- Remove duplicate records
- Convert data types if necessary

3. Exploratory Data Analysis (EDA)

- Analyze booking trends
- Study cancellation patterns
- Identify relationships between variables

4. Data Visualization

- Use charts such as:
  - Strip Plot, Swarm Plot
  - Histogram and Density Plot

- Box Plot and Violin Plot
- Q-Q Plot for normality check
- Correlation Heatmap

5. Tools & Technologies

- Python – Core programming language used for data analysis and machine learning
- Jupyter Notebook / Google Colab – For writing and executing Python code interactively
- Pandas – For data manipulation and preprocessing
- NumPy – For numerical computations and array operations
- Matplotlib – For data visualization and plotting graphs
- Seaborn – For advanced statistical visualizations
- Scikit-learn – For building machine learning models
- Statsmodels – For statistical analysis (e.g., Q-Q plot)
- CSV File (Dataset) – Used as data source
- GitHub – For project hosting and version control (optional)

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8. Expected Outcome

- Predict flight delays accurately
- Identify key factors affecting delays
- Help airlines improve scheduling efficiency
- Improve passenger travel planning

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9. Conclusion

This project demonstrates how data analysis and machine learning techniques can be used to solve real-world aviation problems. It provides insights into flight delay patterns and builds a predictive model that can support decision-making in the airline industry.